

Amitava Shee

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Summary

Results-driven technology leader with 15+ years of experience specializing in building and scaling cloud-native recommendation and search engines at internet scale. Proven track record of delivering high-impact platforms that accelerate time-to-market and achieve significant cost savings. Experienced in leading multidisciplinary teams across AI, software engineering, data science, and platform engineering for global brands in music streaming, fintech, and healthcare.

Experience

Consultant, Services & Platform Engineering | SiriusXM | New York, NY | Nov 2022 - Present

- As acting Director of Personalization, engineered and launched a new recommendation engine for 30M consumers on a new streaming platform, achieving a market launch in a record 13 months.
- Optimized cloud costs by 68%, reducing monthly spend from \$1.57M to \$507K without impacting system performance.
- Led a 20-person engineering team to build and operate a machine learning platform for online and batch inference, processing over 1B records daily at scale.
- Architected a resilient, fault-tolerant platform that enabled rapid innovation and delivered new differentiated product offerings.
- Executing the delivery of MCP-connected apps for ChatGPT, Claude, and Gemini after successful proofs of concept, including AI/LLM and retrieval-augmented generation implementations using PostgreSQL, pgvector, and Neptune.

Chief Technology Officer | reThought Insurance | Broomfield, CO | May 2021 - Mar 2022

- Led engineering teams across Software Development (9), Data Science (3), and DevOps (3) to build, operationalize, and support a new flood insurance underwriting platform.
- Designed and implemented a cloud-native platform on a self-hosted Kubernetes cluster, integrating with multiple third-party API services and risk analytics platforms.
- Enhanced risk scoring metrics by integrating flood exposure data from the National Flood Insurance Program (NFIP) and satellite data providers.

Principal Engineer | AI & Machine Learning | Ithaca | Ann Arbor, MI | Jan 2017 - May 2021

- Drove product growth by implementing advanced search features, including a Learning to Boost (L2B) ML model to dynamically generate search engine parameters.
- Developed a continuous MLOps pipeline to data mine clickstream data, train models, and deploy them to production, accelerating the release cycle.
- Led the migration of distributed search from DataStax to Apache Solr on AWS EKS, resulting in significant cost savings and improved scalability at internet scale.
- Implemented a statistical A/B testing framework to validate product changes and guide data-driven decisions.

Architect, Research Program | University of Michigan Medical School | Ann Arbor, MI | Nov 2013 - Jan 2017

- Delivered MedBus, a SOA-based data services platform that integrated large-scale health data for research, supporting a national healthcare research network.
- Secured approval for an API management solution by building a strong business case and helping bring IBM's products to the enterprise.
- Contributed to the president's \$100M Data Science Initiative and co-led a national committee that successfully delivered over 2M patient records.

Enterprise Architect, Office of CIO | University of Michigan Health System | Ann Arbor, MI | Mar 2011 - Nov 2013

- Established an Enterprise Architecture program based on TOGAF, resulting in the development of a Healthcare Research IT Strategy and Roadmap.
- Evaluated and implemented the organization's first cloud vendor (AWS), paving the way for future cloud adoption.

Information Architect, Enabling Technology | University of Michigan Medical School | Ann Arbor, MI | 2009 - 2011

- Led an applied research program (ENCORE) for medical education, developing rapid applications in Ruby on Rails to support educational research initiatives.

Manager, Enterprise Architecture | University of Michigan Health System | Ann Arbor, MI | Jun 2007 - Jan 2009

- Led the development of clinical application architecture standards, successfully acquiring funding for capital projects through proposal creation and presentation.
- Served as a key stakeholder in the selection of the Epic healthcare system.

Architect, Banking & Financial Services | Cognizant Technology Solutions | Indianapolis, IN | 2004 - 2007

- Partnered with Microsoft to implement a SharePoint-based consumer portal that integrates backend mainframe systems to present a unified view.
- Used BizTalk to integrate core banking systems, including PeopleSoft General Ledger and Loan Origination Systems.

Software Architect | CONTEXTWEB | New York, NY | 2002 - 2004

- Developed an Asset Backed Securities (ABS) online platform for the securities group at Royal Bank of Canada.

Software Architect | Techna International | Sunnyvale, CA | 1999 - 2002

- Acted as a consultant for IBM and Stanford School of Business, helping Galileo transition from mainframe to the web and implementing single sign-on (SSO) solutions.

Senior Software Engineer | CA Technologies | Kolkata, India | 1999 - 1999

- Developed a C++-based Microsoft COM/DCOM integration module in an object database product.

Senior Software Engineer | Tata Elxsi | Rockville, MD | 1995 - 1997

- Provided Electronic Data Interchange (EDI) consulting services for GE Information Services.

Management Trainee | Fujitsu ICIM | Pune, India | 1994 - 1995

- Developed system code in Unix and C/C++.

Skills

Engineering Leadership: Led teams of 20+ engineers to deliver a new streaming platform in a record 13 months, enabling business innovation at scale.

Data & AI Platforms: Built and scaled recommendation and search engines, and led the development of MLOps pipelines for continuous model delivery.

Cloud & Architecture: Architected and modernized cloud-native platforms on AWS and Kubernetes, driving significant cost savings and operational efficiency.

Product Innovation: Drove product growth by implementing advanced search features and building applications with rapid turnaround times.

Enterprise Strategy: Established an Enterprise Architecture program and served as a key stakeholder in the selection of a new healthcare system.

Education

Birla Institute of Technology | B.Eng. Computer Science

Certifications

- MOR Leaders Program | 2016
- Scrum Alliance, License: 000339704 | 2016